

NOTE ON THE IMPLEMENTATION AND EXAMPLES

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1. NOTE ON THE IMPLEMENTATION.

1.1. Recalling the setting. In this subsection, we recall briefly the setting of [CN26]. We keep the same notation.

Let F be a real quadratic field, p a prime number inert in F and ℓ a prime different from p . Attached to $\omega \in F$ and ℓ is a p -unit $u_\ell t(\omega)$ that lives in the narrow ring class field H_ω over F . We recall the generating series involve to compute $u_\ell t(\omega)$ as explains in [CN26] Section 3.1.

Let

$$f_x(z) = \frac{e^{xz}}{e^x - 1} = \sum_{k \geq 0} b_k(z) x^{k-1}.$$

For any $z \in \mathbb{R}$, define its fractional part $\{z\}$ by

$$0 \leq \{z\} < 1 \text{ and } z - \{z\} \in \mathbb{Z}.$$

Let $c > 0, d$ be two coprime integers and $v = (v_1, v_2) \in \mathbb{Q}^2$, we now define the generating series in two variables:

$$(1) \quad g(c, d, v)(x, y) = \sum_{j=0}^{c-1} f_x \left(\left\{ v_1 - \frac{d}{c}(j + v_2) \right\} \right) f_y \left(\frac{j + \{v_2\}}{c} \right) \in \frac{1}{xy} \mathbb{Q}[[x, y]].$$

Let ω be a real quadratic number and $\beta = c\omega + d$, we also define

$$h_\omega(t, c, d, v) = g(c, d, v)(t, \beta t) \in \frac{1}{t^2} \mathbb{Q}(\omega)[[t]].$$

Suppose that c is divisible by an odd prime ℓ , define the smoothed version of h_ω :

$$(2) \quad h_\omega^{(\ell)}(t, c, d, v) := h_{\ell\omega} \left(\frac{t}{\ell}, \frac{c}{\ell}, d, (\ell v_1, v_2) \right) - \ell h_\omega(t, c, d, (v_1, v_2)).$$

Let $p \neq \ell$ be an odd prime number and define a measure $\mu_\ell(c, d)$ on $\mathbb{Z}_p^2 - p\mathbb{Z}_p^2$ as follows. For any $(a, b) \in \mathbb{Z}_p^2 - p\mathbb{Z}_p^2$ and $r \geq 1$, we define the measure on the compact open

$$(3) \quad U_{a,b,r} = (a, b) + p^r \mathbb{Z}_p^2$$

as

$$(4) \quad \mu_\ell(c, d)(U_{a,b,r}) := h_\omega^{(\ell)} \left(t, c, d, \left(\frac{a}{p^r}, \frac{b}{p^r} \right) \right) [t^0].$$

We can write explicitly the right-hand side of Equation (4) as Dedekind sums:

$$(5) \quad \mu_\ell(c, d)(U_{a,b,r}) = \sum_{j=0}^{c/\ell-1} \tilde{b}_1 \left(\ell v_1 - d \frac{(j + v_2)}{c/\ell} \right) \tilde{b}_1 \left(\frac{j + v_2}{c/\ell} \right) - \ell \sum_{j=0}^{c-1} \tilde{b}_1 \left(v_1 - d \frac{(j + v_2)}{c} \right) \tilde{b}_1 \left(\frac{j + v_2}{c} \right),$$

where $\tilde{b}_1$ is the 1-periodic function equal to b_1 on $[0, 1[$ and $(v_1, v_2) = \left(\frac{a}{p^r}, \frac{b}{p^r} \right)$.

We give in [CN26] Section 3.1 formulas to compute $\log_p(u'_\ell(\omega))$, $\text{ord}_p(u'_\ell(\omega))$ and $\log_{\beta_p}(u'_\ell(\omega))$ with M p -adic precision using the coefficients of $h_\omega^{(\ell)}$. Let $\gamma_\omega = \begin{pmatrix} * & * \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ be a generator of $\text{stab}_{SL_2(\mathbb{Z})}(\omega)$ (we choose ω such that $\ell|c$). First, we have

$$(6) \quad \log_p(u'_\ell(\omega)) = 12 \int_{\mathbb{Z}_p^2 - p\mathbb{Z}_p^2} \log_p(x + \omega y) d\mu_\ell(c, d) = 12(C_{\omega, c, d} + I') + O(p^{M+1}),$$

where

$$C_{\omega, c, d} := \sum_{\substack{a, b \pmod p \\ a, b \neq 0, 0}} \log_p(a + b\omega) \mu_\ell(c, d)(U_{a, b, 1}),$$

and

$$(7) \quad I' = \sum_{\substack{a, b \pmod p \\ a, b \neq 0, 0}} \sum_{n=1}^{M'} \frac{1}{n} \sum_{j=0}^n \binom{n}{j} \frac{(-1)^{j-1}}{(a + b\omega)^j} j! p^j h_\omega^{(\ell)}(t, c, d, (\frac{a}{p}, \frac{b}{p})) [t^j] + O(p^{M+1}).$$

Let β_p be a primitive $p^2 - 1$ roots of unity in $K_p = K \otimes \mathbb{Q}_p$ and $\log_{\beta_p}$ be the discrete logarithm with base β_p

$$\log_{\beta_p} : K_p^\times \rightarrow \mathbb{Z}/(p^2 - 1)\mathbb{Z},$$

such that

$$\frac{x}{\beta_p^{\log_{\beta_p}(x)} p^{\text{ord}_p(x)}} \in 1 + p\mathcal{O}_p \text{ for all } x \in K_p^\times.$$

Then, using [Das07] Equation (31) and (44) we have

$$(8) \quad \log_{\beta_p}(u'_\ell(\omega)) = 12 \int_{\mathbb{Z}_p^2 - p\mathbb{Z}_p^2} \log_{\beta_p}(x + \omega y) d\mu_\ell(c, d).$$

By splitting the integral, we obtain

$$(9) \quad \log_{\beta_p}(u'_\ell(\omega)) = 12 \sum_{\substack{a, b \pmod p \\ a, b \neq 0, 0}} \log_{\beta_p}(a + b\omega) h_\omega^{(\ell)}\left(t, c, d, \left(\frac{a}{p}, \frac{b}{p}\right)\right) [t^0].$$

Similarly, we have

$$(10) \quad \text{ord}_p(u'_\ell(\omega)) = 12(h_\omega^{(\ell)}(t, c, d, 0, 0)[t^0] + \frac{\ell - 1}{4}).$$

1.2. Normalization of $\tilde{b}_1$. Equations (9) and (10) are true up to the normalization of $\tilde{b}_1$ at 0. The normalization in [Das07] is $\tilde{b}_1(0) = 0$. To match this normalization to compute $\text{ord}_p(u'_\ell(\omega))$ and $\log_{\beta_p}(u'_\ell(\omega))$, all we have to do is change $f_x(z)$ in the computations by

$$f_x^\#(z) = \begin{cases} \frac{e^{xz}}{e^x - 1} & \text{if } z \neq 0, \\ \frac{e^{xz}}{e^x - 1} + \frac{1}{2} & \text{else.} \end{cases}$$

Algorithm 1 given in [CN26] to compute the coefficients of h_ω works as well.

Furthermore, by taking $\ell \geq 5$ in the smoothing, we can drop the factor 12 in the computations (that is, computing $u'_\ell(\omega)^{1/12}$ instead) as it is the case in [FL21]. When $\ell = 3$ we multiply by 3 instead.

1.3. Cocycle relation. In the computation of the last subsection we see that the running time of the code is linear in c , the lower left coefficient of the stabilizer in $SL_2(\mathbb{Z})$ of ω . As c can become quite large, we explain how to get rid of this dependence.

Using the notations of [Das07], we have

$$(11) \quad \mu_\ell(c, d) = \mu_\ell\left(\infty \rightarrow \frac{c}{d}\right).$$

For all compact open $U \subset \mathbb{Z}_p^2 - p\mathbb{Z}_p^2$ and $\gamma \in \Gamma_0(\ell)$, we have

$$(12) \quad \mu_\ell(\gamma r \rightarrow \gamma s)(\gamma U) = \mu_\ell(r \rightarrow s)(U).$$

As in [Das07] Algorithm 4.1 and [FL21], we write

$$(13) \quad \left(\infty \rightarrow \frac{c}{d}\right) = \sum_{\substack{i=1..\ell-1 \\ \gamma \in \Gamma_0(N)}} a_{\gamma,i} \gamma \left(\infty \rightarrow \frac{i}{\ell}\right)$$

where almost all $a_{\gamma,i}$ are zero. In particular, we can rewrite the integral giving $\log_p(u'_\ell(\omega))$ as

$$(14) \quad \int_{\mathbb{Z}_p^2 - p\mathbb{Z}_p^2} \log_p(x + \omega y) d\mu(c, d) = \sum_{i, \gamma} a_{\gamma,i} \int_{\mathbb{Z}_p^2 - p\mathbb{Z}_p^2} \log_p(x - y\gamma^{-1}(-\omega)) d\mu(\ell, i),$$

see [Das07] Equation (45). We use Equation (6) to compute these integrals. We use the same decomposition to compute $\log_\beta(u'_\ell(\omega))$ and $\text{ord}_p(u'_\ell(\omega))$. See [FL21] for more details.

2. NUMERICAL EXAMPLES

The computations of this section were done on the Team Ouragan (INRIA, IMJ-PRG) servers with 128GB of RAM and 32 Intel i9-13900 processors. The code was run using SageMath version 9.5 [The22].

The general strategy to compute $P_{u'_\ell(\omega)}$ is as follows. For each class in the narrow class group of F , we find a quadratic form $a_ix^2 + b_i xy + c_i y^2$ with $\ell|a_i$. We pick $\omega_i = \frac{-b_i + \sqrt{D}}{2a_i}$, the minimal polynomial over F is then given by

$$P_{u_\ell(\omega)} = P_{u'_\ell(\omega)} = \prod_i (x - u_\ell(\omega_i)).$$

Example 1. Let $D = 24$ and $\ell = 5$, we take

$$\omega = \frac{\sqrt{D}}{10} + \frac{4}{5}, \quad \gamma_\omega = \begin{pmatrix} 13 & -4 \\ 10 & -3 \end{pmatrix}.$$

In particular, we have $(c, d) = (10, -3)$. We compute for the following primes the unit $u_\ell(-\omega)$ with precision $M = 100$. As the narrow class group of $F = \mathbb{Q}(\sqrt{24})$ is of order 2, we know that the minimal polynomial $P_{u'_\ell(\omega)}$ of $u'_\ell(\omega)$ with coefficients in K is of degree 2. In particular

$$P_{u_\ell(\omega)} = (x - u_\ell(\omega))(x - u'_\ell(\omega)) = (x - u_\ell(\omega))(x - u_\ell(\omega)^{-1}).$$

We use Sagemath to recognize the coefficient of x of $P_{u_\ell(\omega)}$. We have the following table.

p	$P_{u_\ell(\omega)}$	Computation time
7	$x^2 - \frac{11}{7}x + 1$	55s
11	$x^2 + \left(-\frac{3}{11}\sqrt{D} - \frac{7}{11}\right)x + 1$	2m 30s
13	$x^2 - \frac{1}{13}x + 1$	3m 37s
17	$x^2 + \left(-\frac{6}{17}\sqrt{D} + \frac{1}{17}\right)x + 1$	6m 39s
31	$x^2 + \frac{13}{31}x + 1$	25m 15s
37	$x^2 + \frac{26}{37}x + 1$	38m 12s
41	$x^2 + \left(\frac{12}{41}\sqrt{D} + \frac{23}{41}\right)x + 1$	45m 9s

TABLE 1. Minimal polynomial for $D = 24$, $\ell = 5$ and $M = 100$

As the prime p grows, we reduce the precision M that we need to recognize the coefficients as elements of $\mathbb{Q}(\omega)$.

p	$P_{u_\ell(\omega)}$	Computation time
59	$x^2 + \frac{82}{59}x + 1$	15m 26s
61	$x^2 - \frac{47}{61}x + 1$	14m 1s
79	$x^2 + \frac{142}{79}x + 1$	24m 8s
83	$x^2 + \left(\frac{9}{83}\sqrt{D} - \frac{79}{83}\right)x + 1$	26m 49s
89	$x^2 + \left(-\frac{18}{89}\sqrt{D} + \frac{73}{89}\right)x + 1$	31m 17s
103	$x^2 - \frac{194}{103}x + 1$	42m 8s
107	$x^2 + \left(-\frac{21}{107}\sqrt{D} - \frac{89}{107}\right)x + 1$	45m 37s
109	$x^2 - \frac{143}{109}x + 1$	47m 55s

TABLE 2. Minimal polynomial for $D = 24$, $\ell = 5$ and $M = 50$

p	$P_{u_\ell(\omega)}$	<i>Computation time</i>
521	$x^2 + \left(\frac{48}{521}\sqrt{D} - \frac{503}{521}\right)x + 1$	37m 5s
541	$x^2 - \frac{793}{541}x + 1$	39m 49s
563	$x^2 + \frac{226}{563}x + 1$	46m 30s
569	$x^2 + \left(-\frac{168}{569}\sqrt{D} + \frac{313}{569}\right)x + 1$	47m 43s
587	$x^2 + \left(\frac{51}{587}\sqrt{D} + \frac{569}{587}\right)x + 1$	50m 40s

TABLE 3. Minimal polynomial for $D = 24$, $\ell = 5$ and $M = 10$

p	$P_\ell(u_\ell(\omega))$	<i>Computation time</i>
7	$x^2 - \frac{11}{7}x + 1$	16h 40m 15s
11	$x^2 + \left(-\frac{3}{11}\sqrt{D} - \frac{7}{11}\right)x + 1$	47h 59m 45s

TABLE 4. Minimal polynomial for $D = 24$, $\ell = 5$, $M = 1000$

We see in Table 3 that we are able to compute the values for large p . Our algorithm also allows us to work with very high p -adic precision when p is small. In Table 4, we give examples with 1000 p -adic digit accuracy, a comparable precision to similar computations done in the archimedean world see for example [BCG23].

The polynomials in these tables correspond to the same extension H of $F = \mathbb{Q}(\sqrt{24})$. Here

$$(15) \quad H = F(j) = \mathbb{Q}(\sqrt{24}, j) = \mathbb{Q}(\sqrt{24} + j),$$

where $j = e^{\frac{2i\pi}{3}}$. The field H is, as expected, the narrow class field of F .

Example 2. Let $D = 156$ and $\ell = 5$, take

$$(16) \quad \omega_1 = \frac{\sqrt{D}}{20} + \frac{3}{10}, \quad \gamma_{\omega_1} = \begin{pmatrix} 37 & 12 \\ 40 & 13 \end{pmatrix}, \quad \omega_2 = \frac{\sqrt{D}}{10} - \frac{2}{5}, \quad \gamma_{\omega_2} = \begin{pmatrix} 17 & 28 \\ 20 & 33 \end{pmatrix}.$$

This time we have $(c_1, d_1) = (40, 13)$ and $(c_2, d_2) = (20, 33)$. As the computation time is linear in c , the computations for this examples should be six times longer. The narrow class group of $F = \mathbb{Q}(\sqrt{156})$ is of degree 4, $u_\ell(\omega_1)$ and $u_\ell(\omega_2)$ are conjugate, their minimal polynomial over F is given by

$$P_\omega = (x - u_\ell(\omega_1))(x - u_\ell(\omega_1)^{-1})(x - u_\ell(\omega_2))(x - u_\ell(\omega_2)^{-1}) \in K[x].$$

The polynomial $P_{u_\ell(\omega)}$ is palindromic, we write only the first coefficients.

p	$P_{u_\ell(\omega)}$	Computation time
11	$x^4 + \left(-\frac{24}{11^3}\sqrt{D} - \frac{2808}{11^3}\right)x^3 + \left(\frac{6}{11^2}\sqrt{D} + \frac{36137}{11^4}\right)x^2 + ..$	8m 56s
17	$x^4 + \left(-\frac{644}{17^3}\sqrt{D} + \frac{2898}{17^3}\right)x^3 + \left(-\frac{36}{17^2}\sqrt{D} + \frac{140387}{17^4}\right)x^2 + ..$	21m 28s

TABLE 5. Minimal polynomial for $D = 156$, $\ell = 5$ and $M = 100$

p	$P_{u_\ell(\omega)}$	Computation time
73	$x^4 - \frac{66332}{73^3}x^3 - \frac{43440954}{73^4}x^2 + ..$	1h 29m 37s
79	$x^4 + \frac{1417444}{79^3}x^3 + \frac{147583686}{79^4}x^2 + ..$	1h 44m 32s
83	$x^4 + \left(\frac{3128}{83^3}\sqrt{D} + \frac{150696}{83^3}\right)x^3 + \left(\frac{714}{83^2}\sqrt{D} + \frac{32851337}{83^4}\right)x^2 + ..$	1h 56m 8s
97	$x^4 + \frac{324142}{97^3}x^3 + \frac{95350611}{97^4}x^2 + ..$	2h 38m 28s
101	$x^4 + \left(-\frac{263228}{101^3}\sqrt{D} + \frac{56406}{101^3}\right)x^3 + \left(-\frac{84}{101^2}\sqrt{D} + \frac{457391987}{101^4}\right)x^2 + ..$	2h 53m 57s
103	$x^4 - \frac{3631292}{103^3}x^3 + \frac{524542086}{103^4}x^2 + ..$	2h 59m 50s
109	$x^4 + \frac{1338766}{109^3}x^3 + \frac{48285291}{109^4}x^2 + ..$	3h 25m 19s

TABLE 6. Minimal polynomial for $D = 156$, $\ell = 5$ and $M = 50$

p	$P_{u_\ell(\omega)}$	Computation time
521	$x^4 + \frac{59731644}{521^3}x^3 - \frac{75691649914}{521^4}x^2 + ..$	2h 53m 37s
523	$x^4 + \frac{43862518}{523^3}x^3 - \frac{80826503229}{523^4}x^2 + ..$	2h 40m 43s
541	$x^4 + \frac{95348734}{541^3}x^3 + \frac{27184995291}{541^4}x^2 + ..$	2h 49m 45s
547	$x^4 + \frac{254172908}{547^3}x^3 + \frac{149879597286}{547^4}x^2 + ..$	2h 55m 42s
569	$x^4 + \left(\frac{32989928}{569^3}\sqrt{D} + \frac{66413934}{569^3}\right)x^3 + \left(\frac{23256}{569^2}\sqrt{D} + \frac{278087052707}{569^4}\right)x^2 + ..$	3h 8m 9s
571	$x^4 - \frac{593232794}{571^3}x^3 + \frac{467704609011}{571^4}x^2 + ..$	3h 10m 6s
577	$x^4 + \frac{46621582}{577^3}x^3 + \frac{211173573171}{577^4}x^2 + ..$	3h 15m 21s

TABLE 7. Minimal polynomial for $D = 156$, $\ell = 5$ and $M = 10$

All the polynomials in these tables generate the same unramified abelian extension H of F . The field H is generated over $\mathbb{Q}$ by the polynomial

$$(17) \quad \begin{aligned} P(X) = & 214358881 \times x^8 - 904462416 \times x^7 + 2001359602 \times x^6 - 3077062560 \times x^5 \\ & + 3538768611 \times x^4 - 3077062560 \times x^3 + 2001359602 \times x^2 - 904462416 \times x + 214358881. \end{aligned}$$

Example 3. For $D = 12$ and $\ell = 3$ we take

$$(18) \quad \omega = \frac{\sqrt{D}}{6} = \frac{1}{\sqrt{3}}, \quad \gamma_\omega = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}.$$

In particular $c = 3, d = 2$.

p	$P_{u_\ell(\omega)}$	Computation time
5	$x^2 + \frac{8}{5}x + 1$	0m 2s
7	$x^2 + \frac{13}{7}x + 1$	0m 5s
17	$x^2 - \frac{16}{17}x + 1$	0m 34s
19	$x^2 - \frac{11}{19}x + 1$	0m 42s
29	$x^2 + \frac{40}{29}x + 1$	1m 46s
31	$x^2 + \frac{46}{31}x + 1$	1m 56s
41	$x^2 - \frac{80}{41}x + 1$	3m 43s
43	$x^2 - \frac{22}{43}x + 1$	4m 7s
53	$x^2 - \frac{56}{53}x + 1$	6m 19s
67	$x^2 + \frac{109}{67}x + 1$	8m 57s
79	$x^2 + \frac{131}{79}x + 1$	12m 10s
89	$x^2 - \frac{160}{89}x + 1$	15m 53s
101	$x^2 - \frac{40}{101}x + 1$	20m 20s
103	$x^2 - \frac{37}{103}x + 1$	21m 20s

TABLE 8. Minimal polynomial for $D = 12$, $\ell = 3$ and $M = 50$

All these polynomials generate over $\mathbb{Q}(\sqrt{12})$ the narrow Hilbert class field H_{12} generated over $\mathbb{Q}$ by

$$x^4 - 22x^2 + 169$$

p	$P_{u_\ell(\omega)}$	Computation time
1229	$x^2 - \frac{280}{1229}x + 1$	41m 0s
1231	$x^2 - \frac{2101}{1231}x + 1$	40m 32s
1277	$x^2 - \frac{1496}{1277}x + 1$	45m 20s
1279	$x^2 - \frac{709}{1279}x + 1$	47m 8s
1289	$x^2 + \frac{1120}{1289}x + 1$	48m 12s
1291	$x^2 - \frac{1907}{1291}x + 1$	48m 5s
1301	$x^2 + \frac{2600}{1301}x + 1$	49m 17s
1303	$x^2 - \frac{419}{1303}x + 1$	48m 14s
1327	$x^2 - \frac{2638}{1327}x + 1$	50m 0s
1361	$x^2 - \frac{2480}{1361}x + 1$	53m 0s
1373	$x^2 + \frac{296}{1373}x + 1$	52m 4s
1399	$x^2 - \frac{1826}{1399}x + 1$	51m 41s
1409	$x^2 + \frac{2800}{1409}x + 1$	52m 9s
1423	$x^2 + \frac{2446}{1423}x + 1$	53m 18s
1433	$x^2 - \frac{1184}{1433}x + 1$	54m 8s
1447	$x^2 - \frac{1669}{1447}x + 1$	55m 45s
1459	$x^2 - \frac{218}{1459}x + 1$	56m 12s
1471	$x^2 - \frac{2834}{1471}x + 1$	57m 6s
1481	$x^2 + \frac{2240}{1481}x + 1$	57m 25s
1483	$x^2 + \frac{1597}{1483}x + 1$	58m 21s
1493	$x^2 - \frac{1064}{1493}x + 1$	59m 10s
1531	$x^2 - \frac{3013}{1531}x + 1$	1h 2m 26s
1543	$x^2 + \frac{2843}{1543}x + 1$	1h 3m 10s
1553	$x^2 - \frac{2944}{1553}x + 1$	1h 4m 0s
1567	$x^2 - \frac{3107}{1567}x + 1$	1h 5m 18s
1579	$x^2 - \frac{2134}{1579}x + 1$	1h 6m 4s

TABLE 9. Minimal polynomial for $D = 12$, $\ell = 3$ and $M = 5$

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